MIL-M-28787/20A(EC)
22 March 1979
SUPERSEDING
MIL-M-28787/20(NAVY)
31 October 1973

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INVERTER, DIGITAL, KEY CODE LDC

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic inverter module.

1.2 Module description. The following electrical parameters reflect the -55° to 100°C end-of-life limits from table II.

Name	Inverter
Family	Digital
Key code	LDC
Size	1A
Weight	35 grams maximum
Failure rate	0.18 failures/10^6 hours maximum for class I 0.45 failures/10^6 hours maximum for class II
Environment	Classes I and II as specified in MIL-STD-1389
Supply voltage	5.0 Vdc ±10 percent
Maximum power dissipation	
All outputs in low level state	0.545 W
All outputs in high level state	0.198 W
Input current	
Low level (V_{IL} = 0 V)	1.73 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
Input threshold voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Output voltage	
Low level	0.V maximum
High level	2.4 V minimum, V_{CC} maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Output current	
Low level (V_{OL} = 0.4 V)	16 mA maximum
High level (V_{OH} = 2.4 V)	1 mA maximum
Propagation delay time	
t_{PHL}	2 ns minimum, 22 ns maximum
t_{PLH}	4 ns minimum, 30 ns maximum

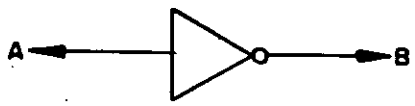
1.3 Module function. This module contains 18 independent inverters.

1.3.1 Input to output relationship. Each inverter has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship.

Input (A)	Output (B)
L	H
H	L

L = Low-voltage level.
H = High-voltage level.



$B = \overline{A}$

1.3.2 Parallel outputs. Paralleling of outputs is not permissible except when both inputs and outputs of circuits are paralleled to obtain extended drive. Each circuit whose output is paralleled with an output of another circuit should have each of its inputs paralleled with a corresponding input of the other circuit. Circuits may be paralleled only with other circuits within the same group. These circuit groups are as follows:

1. Circuits 1,2,3,4,5, and 6.
2. Circuits 7,8,9,10,11, and 12.
   Circuits 13,14,15,16,17,and 18.

The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

1.3.3 Crosstalk relationships. The module has the following crosstalk relationships.

1.3.3.1 Module crosstalk. When connected in the worst case mode of operation (all but one gate switching simultaneously from a high level to a low level), the LDC module could exhibit substantial crosstalk. When the quiet output line is a low level, there may be as much as 970 mV positive going spike coincident with the negative edge of the switching outputs.

1.3.3.2 Circuit crosstalk. The crosstalk on any output pin is nearly directly proportional to the number of simultaneously switching outputs. For every switching output in group A (circuits 1 through 6), the noise induced on a low level quiet output in group A is as much as 110 mV. For every switching gate outside group A (circuits 7 through 18), the noise induced on a low level quiet output in group A is as much as 35 mV. For every switching output in group B (circuits 7 through 12), the noise induced on a low level quiet output in group B is as much as 90 mV. For every switching output outside group B (circuits 1 through 6 and 13 through 18), the noise induced on a low level quiet output in group B is as much as 35 mV. For every switching output in group C (circuits 13 through 18), the noise induced on a low level quiet output in group C is as much as 110 mV. For every switching output outside group C (circuits 1 through 12), the noise induced on a low level quiet output in group C is as much as 35 mV.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.18 for class I and 0.45 for class II failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design classes I and II as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LDC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/20-1 for class I and M28787/20-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

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3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current and output impedance apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	4
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Pulse generator	Item J01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Square wave generator	Item H01	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y02	1
Megohm bridge	Item R06	1
Capacitance meter	Item C05	1
Resistors		
316Ω ±1%, 1/4 W		18
877Ω ±1%, 1/8 W		1
51Ω ±5%, 1 W		2
Value as required for figure 9, note 1, 1/4 W		18
Value as required for figure 9, note 2, 1/4 W		18
511Ω ±1%, 1/4 W		1
10 kΩ ±1%, 1/8 W		1
Capacitors		
100 pF ±20%		18
Value as required for figure 9, note 3		1
Value as required for figure 9, note 4		18
Diode		
1N4150, or equivalent		18

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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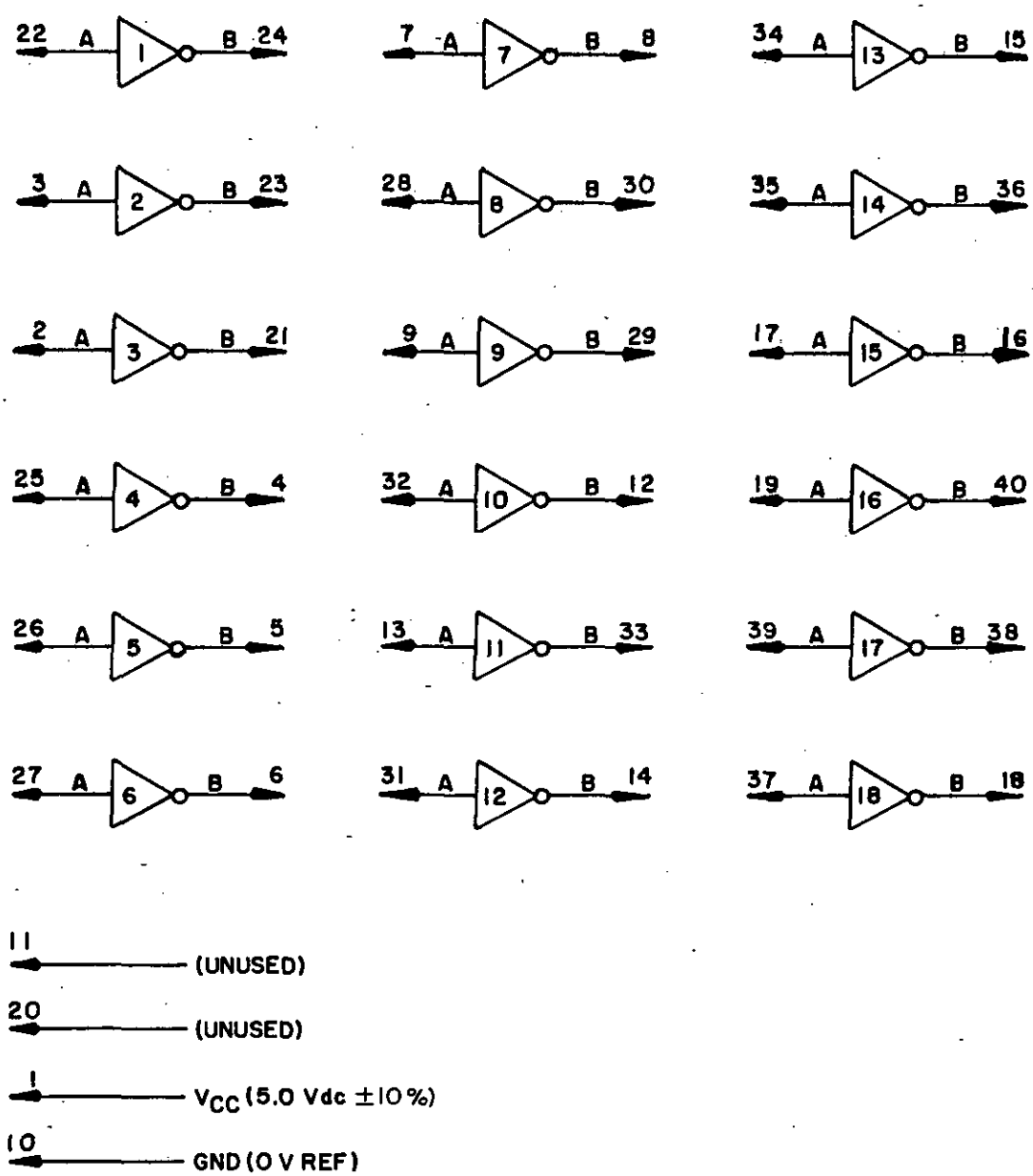


FIGURE 1. LDC module function diagram, inverter.

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TABLE II. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type (see notes)	-55° +5°C Initial & end-of-life limits		0°+5°C Initial & end-of-life limits		25° +5°C Initial & end-of-life limits		60°+5°C Initial & end-of-life limits		100°+5°C Initial & end-of-life limits		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	V_{OUT} (low-loaded)	III	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B1-B18	Input breakdown current	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C1-C18	Input clamp diode voltage	V	Q	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D18	Output voltage, low level	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E18	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F18	Input current, low level	VIII	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
G1-G18	Input current, high level	IX	X	0	40	0	40	0	30	0	40	0	40	µA
H1-H18	Output short circuit current	X	X	20	55	20	55	20	55	20	55	20	55	mA
J1	Power supply current, low level	XI	S	1	99	1	99	1	99	1	99	1	99	mA
J2	Power supply current, high level	XI	S	1	36	1	36	1	36	1	36	1	36	mA
K1-K18	Transition time (t_{TLH})	XII	S	4	50	4	50	4	45	4	50	4	50	ns
L1-L18	Transition time (t_{THL})	XII	S	4	25	4	25	4	20	4	25	4	25	ns
M1-M18	Propagation delay time (t_{PHL})	XIII	S	2	22	2	22	2	19	2	22	2	22	ns

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type (see notes)	-55° +5°C Initial & end-of-life limits		0°+5°C Initial & end-of-life limits		25° +5°C Initial & end-of-life limits		60°+5°C Initial & end-of-life limits		100°+5°C Initial & end-of-life limits		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1-N18	Propagation delay time (t_{PLH})	XIII	S	4	30	4	30	4	28	4	30	4	30	ns
P1-P6, P13-P18	Output crosstalk, low level	XIV	Q	0	970	0	970	0	970	0	970	0	970	mV
P7-P12	Output crosstalk, low level	XIV	Q	0	870	0	870	0	870	0	870	0	870	mV
P19-P36	Output crosstalk, low level	XIV	Q	0	420	0	420	0	420	0	420	0	420	mV
R1-R18	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVI	Q	0	600	0	600	0	600	0	600	0	600	mV
T	Power supply transient rise time	XVII	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
U1	Isolation	XVIII	S											
	Initial			10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life			1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
U2, U3	Isolation, unused pins	XVIII	Q											
	Initial			10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life			1	∞	1	∞	1	∞	1	∞	1	∞	MΩ

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type (see notes)	-55° +5°C Initial & end-of-life limits		0°+5°C Initial & end-of-life limits		25° +5°C Initial & end-of-life limits		60°+5°C Initial & end-of-life limits		100°+5°C Initial & end-of-life limits		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
V1-V18	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF
W1-W36	Output impedance, power off, check	XX	Q	3.5	5.55	3.5	5.55	3.5	5.55	3.5	5.55	3.5	5.55	V

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.
4. Unless otherwise specified, limits are for initial and end-of-life.

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4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, or U are part of a sequence of tests, the order of testing shall be as follows: A, B, U where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through N, and T, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test condition T, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through R, and T, shall be performed in the order specified in 4.4.2 using the maximum class temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests C through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

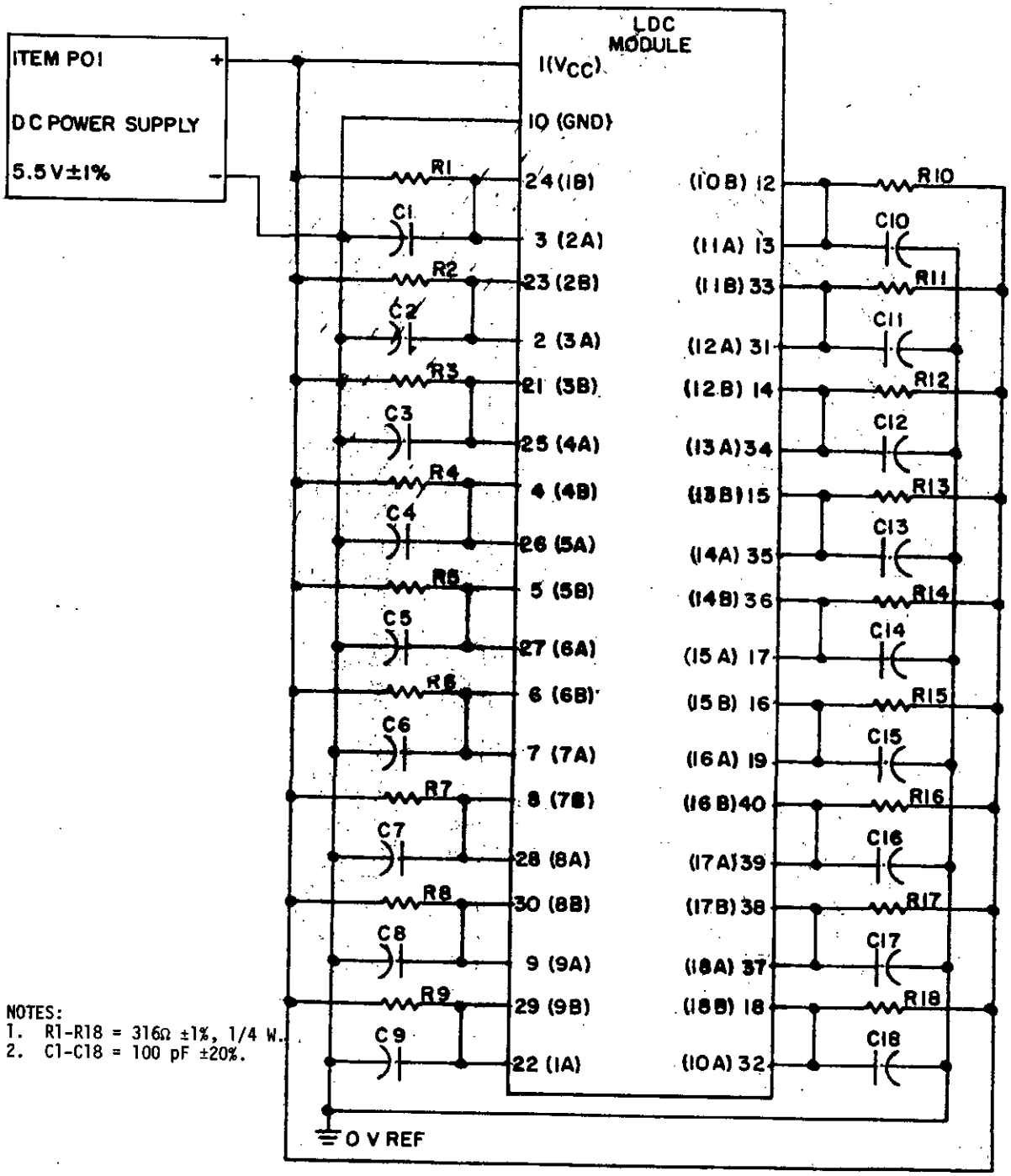
6.3 Cross reference of classification. Part number M28787/20-1 supersedes part number M28787/20. The two parts are interchangeable.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N024)

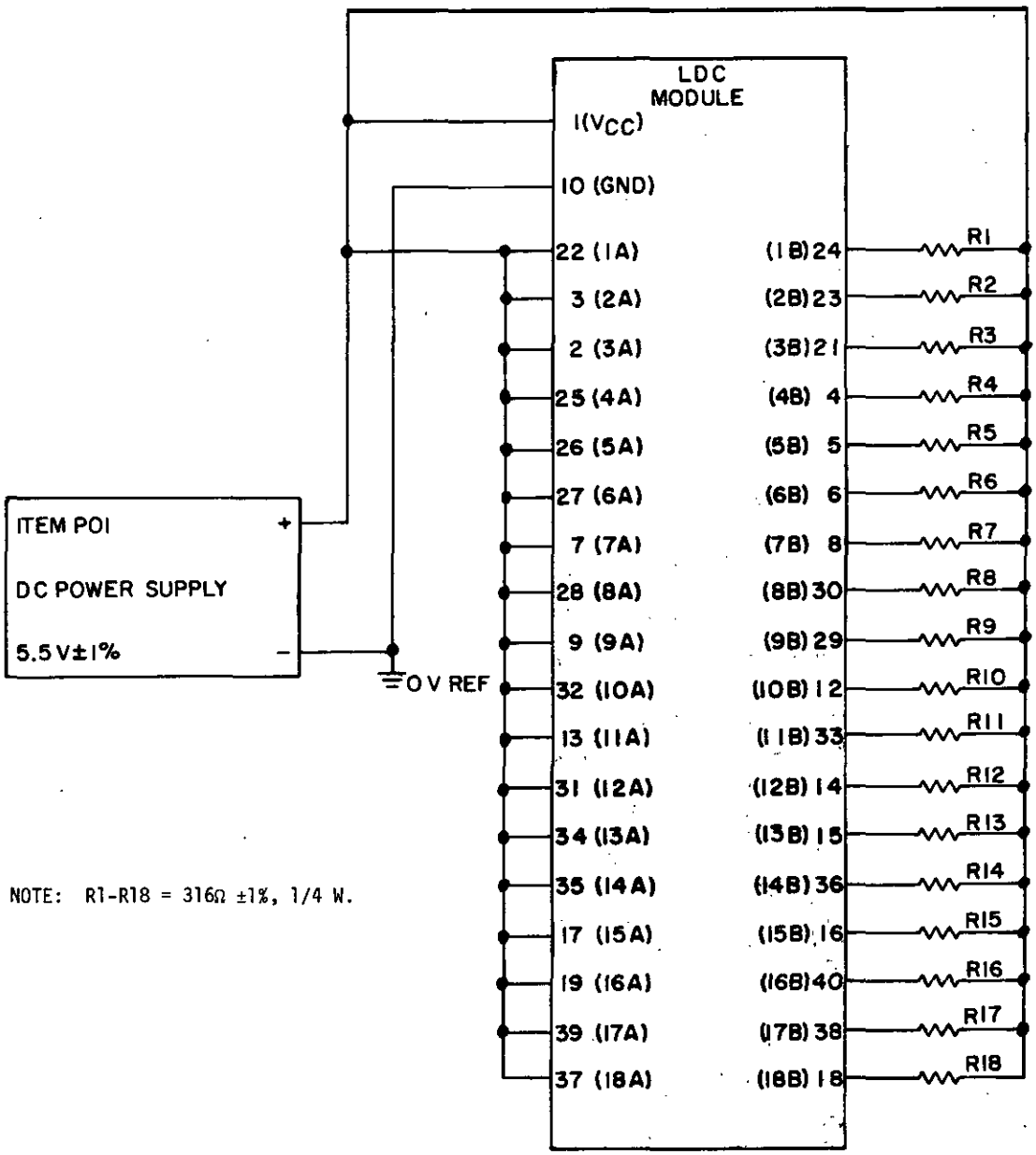
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NOTES:
1. R1-R18 = 316Ω ±1%, 1/4 W.
2. C1-C18 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R18 = 316Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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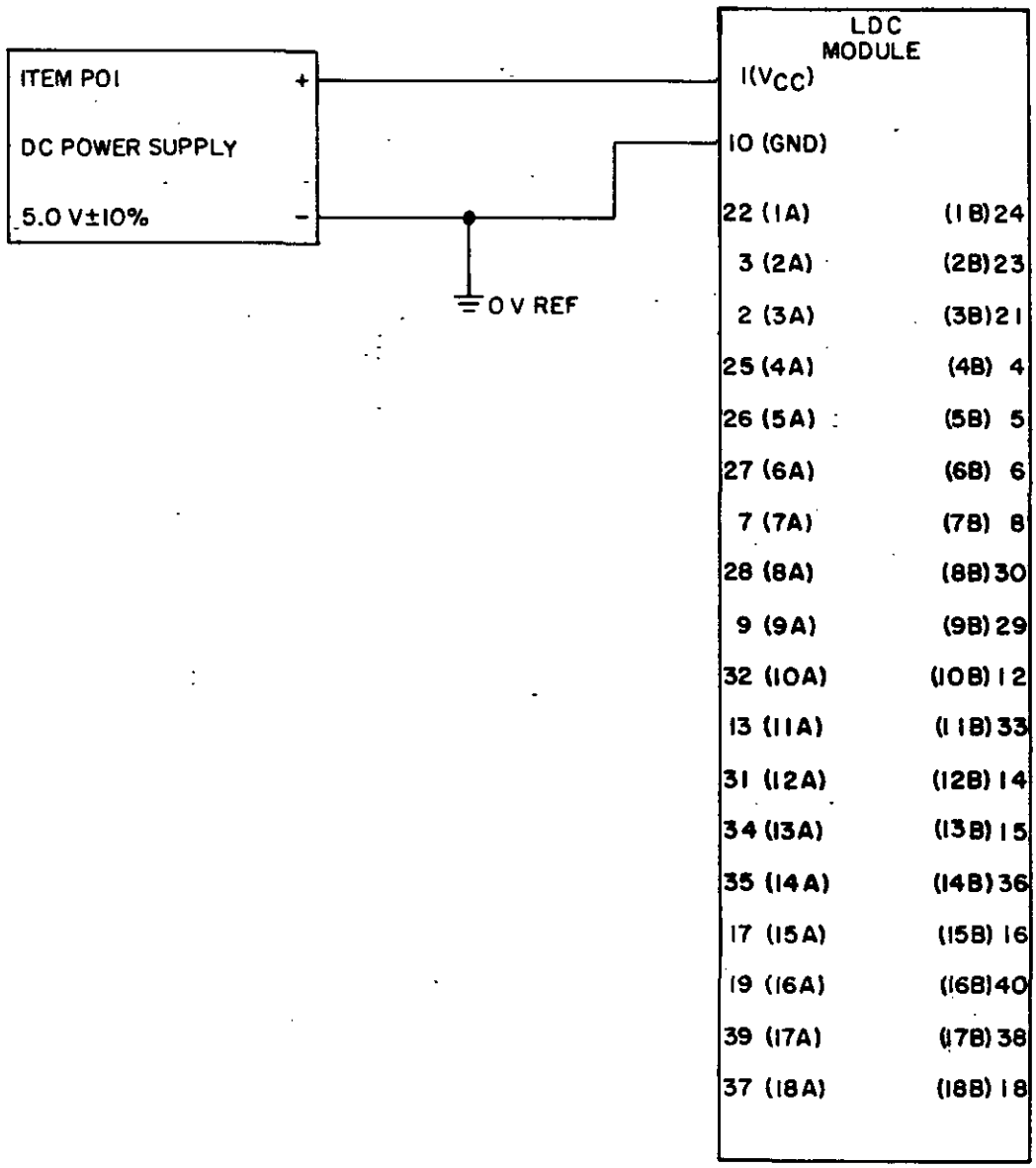


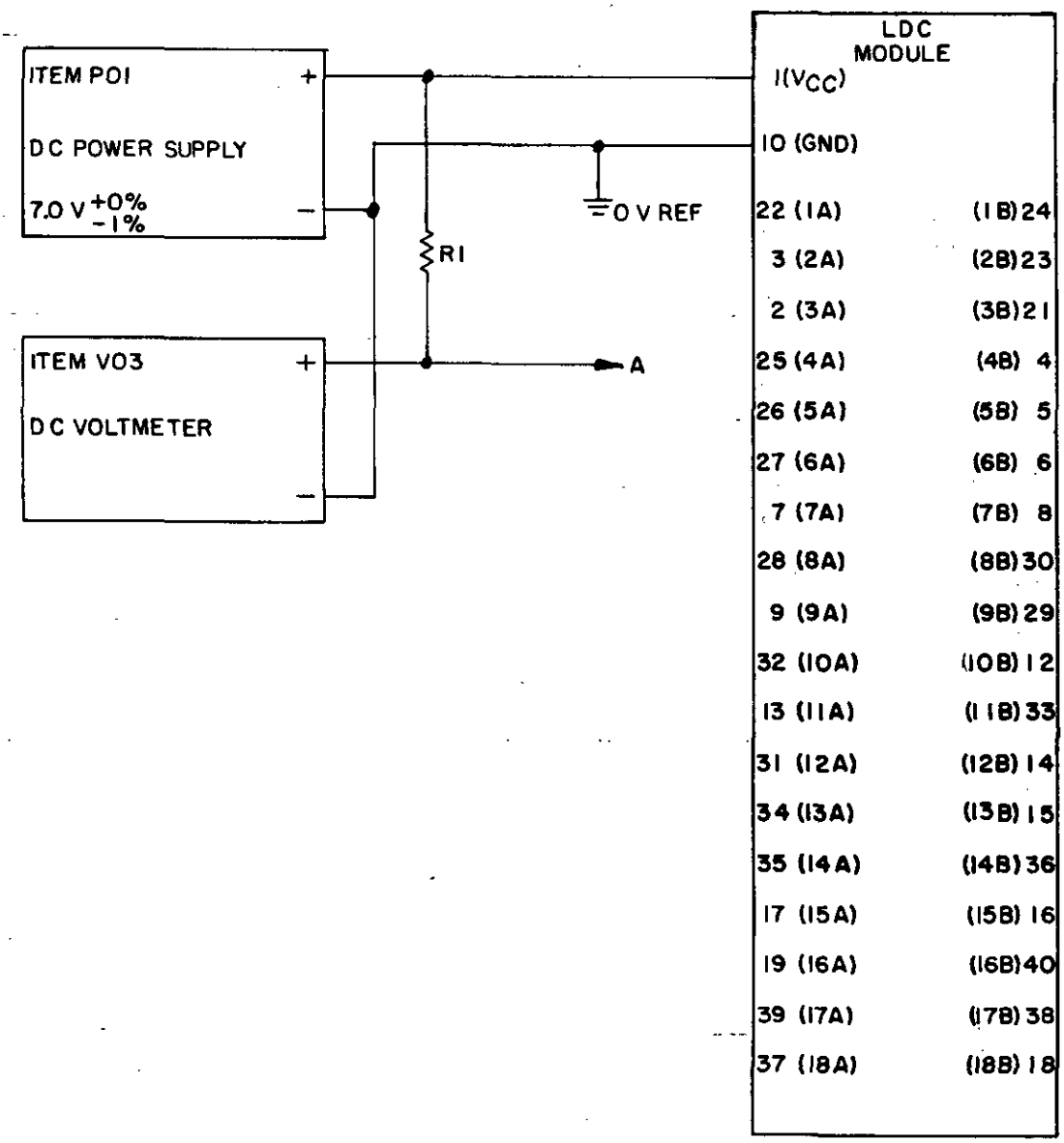
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
A1	B1 V_{OUT} (low loaded)	24	0.01	0.8	V	5
A2	B2	23				
A3	B3	21				
A4	B4	4				
A5	B5	5				
A6	B6	6				
A7	B7	8				
A8	B8	30				
A9	B9	29				
A10	B10	12				
A11	B11	33				
A12	B12	14				
A13	B13	15				
A14	B14	36				
A15	B15	16				
A16	B16	40				
A17	B17	38				
A18	B18 V_{OUT} (low loaded)	18	0.01	0.8	V	5

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NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. Measurements shall be recorded after point A has been connected to the appropriate module pin for at least 5 seconds.

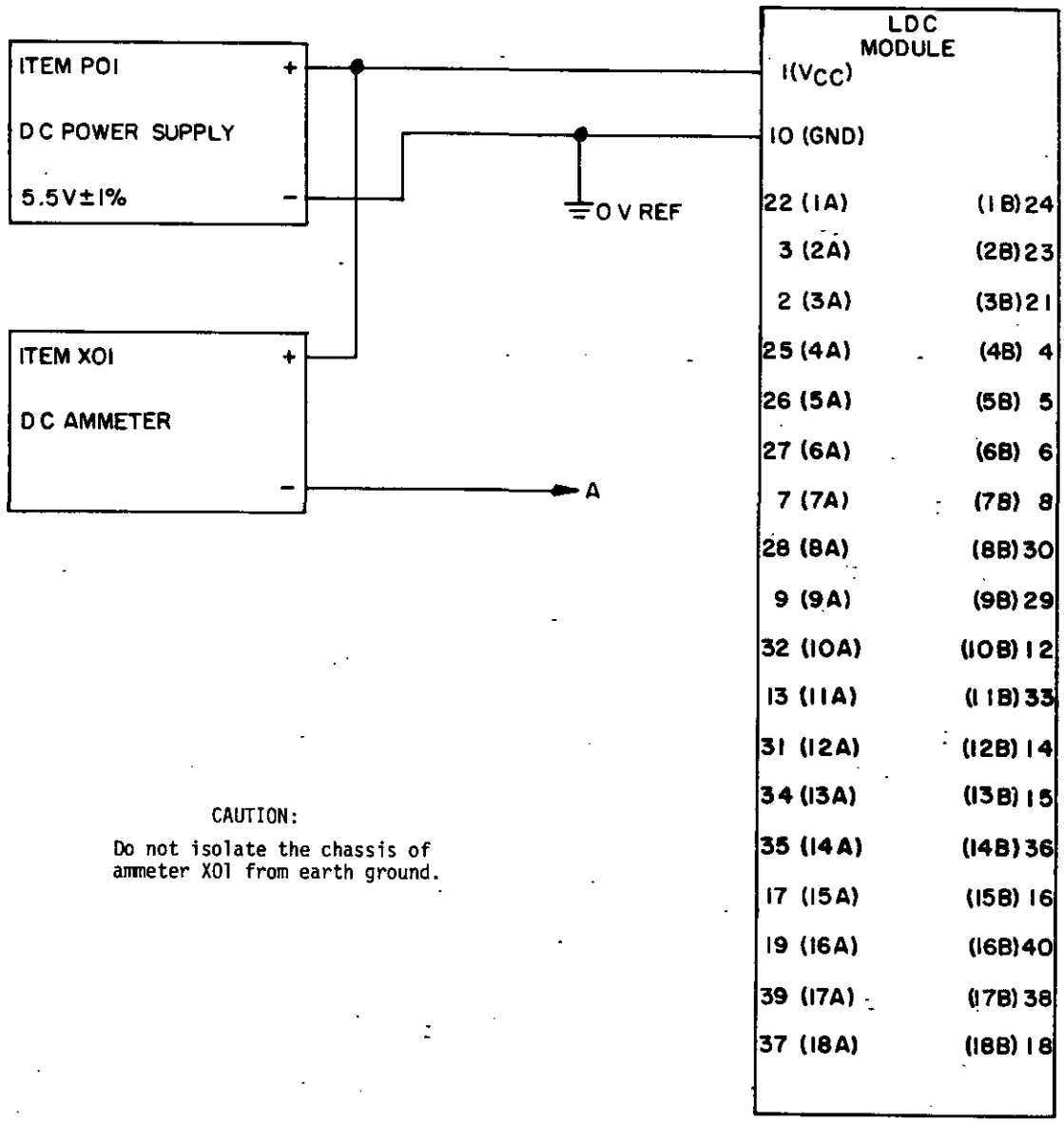
FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition		Parameter	Connect	Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
B1	A1	Input breakdown current	22	0.0	1.0	mA	6
B2	A2		3				
B3	A3		2				
B4	A4		25				
B5	A5		26				
B6	A6		27				
B7	A7		7				
B8	A8		28				
B9	A9		9				
B10	A10		32				
B11	A11		13				
B12	A12		31				
B13	A13		34				
B14	A14		35				
B15	A15		17				
B16	A16		19				
B17	A17		39				
B18	A18	Input breakdown current	37	0.0	1.0	mA	6

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

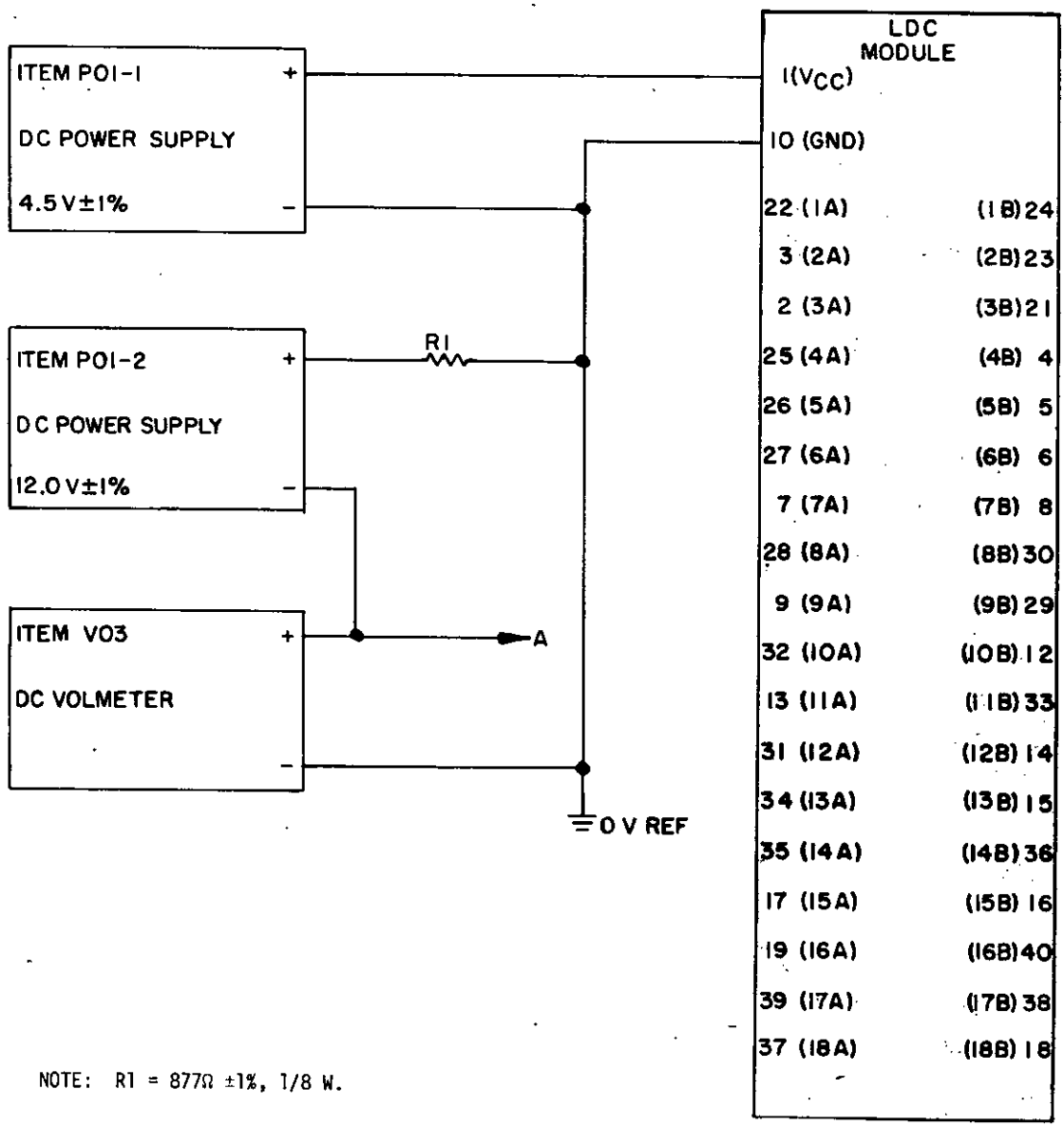
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

Test condition		Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
C1	A1	Input clamp diode voltage	22	-0.4	-1.5	V	7
C2	A2		3				
C3	A3		2				
C4	A4		25				
C5	A5		26				
C6	A6		27				
C7	A7		7				
C8	A8		28				
C9	A9		9				
C10	A10		32				
C11	A11		13				
C12	A12		31				
C13	A13		34				
C14	A14		35				
C15	A15		17				
C16	A16		19				
C17	A17		39				
C18	A18	Input clamp diode voltage	37	-0.4	-1.5	V	7

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NOTE: R1 = 877Ω ±1%, 1/8 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

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TABLE VI. Output voltage (low-level) test.

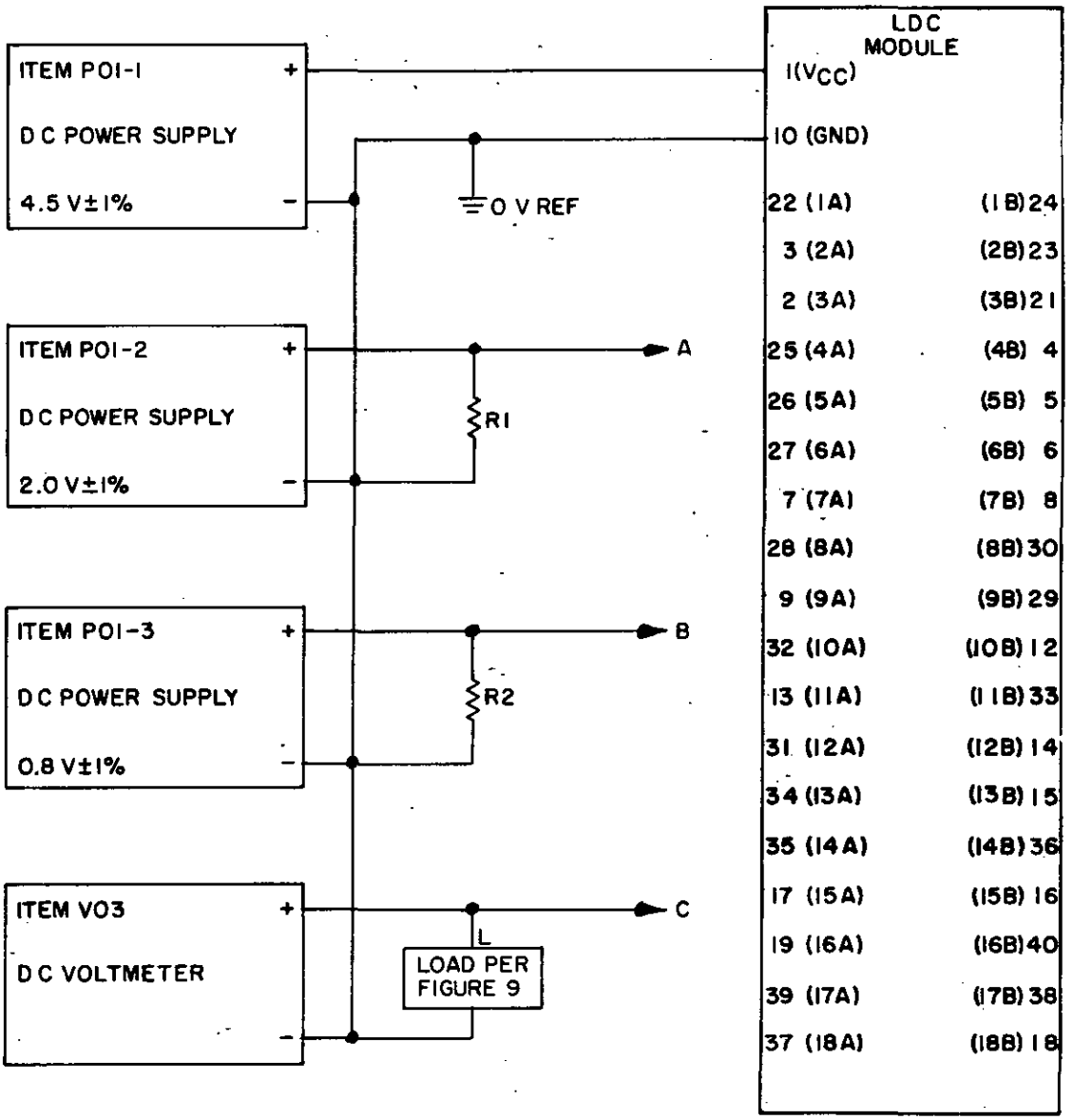
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	B1	Output voltage, low level	22	Open	24	0.0	0.4	V	8
D2	B2		3		23				
D3	B3		2		21				
D4	B4		25		4				
D5	B5		26		5				
D6	B6		27		6				
D7	B7		7		8				
D8	B8		28		30				
D9	B9		9		29				
D10	B10		32		12				
D11	B11		13		33				
D12	B12		31		14				
D13	B13		34		15				
D14	B14		35		36				
D15	B15		17		16				
D16	B16		19		40				
D17	B17		39		38				
D18	B18	Output voltage, low level	37	Open	18	0.0	0.4	V	8

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TABLE VII. Output voltage (high-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	B1	Output voltage, high level	Open	22	24	2.4	4.5	V	8
E2	B2			3	23				
E3	B3			2	21				
E4	B4			25	4				
E5	B5			26	5				
E6	B6			27	6				
E7	B7			7	8				
E8	B8			28	30				
E9	B9			9	29				
E10	B10			32	12				
E11	B11			13	33				
E12	B12			31	14				
E13	B13			34	15				
E14	B14			35	36				
E15	B15			17	16				
E16	B16			19	40				
E17	B17			39	38				
E18	B18	Output voltage, high level	Open	37	18	2.4	4.5	V	8

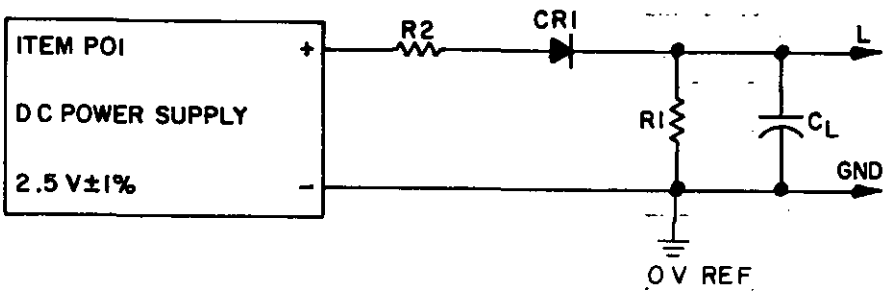
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NOTE: R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 8. Output voltage (low-level and high-level) test circuit for tables VI and VII.

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NOTES:
1. R1 = Select value for a current flow of 1.0 mA ±1% into point L when point L is held at 2.4 Vdc.
2. R2 = Select value for a current flow of 17.3 mA ±1% from point L to ground.
3. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions K, L, M, and N.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P and R.
5. CR1 = 1N4150, or equivalent.
6. C_L is not required for dc electrical tests D and E.
7. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
8. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 9. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	A1	Input current, low level	Open	22	Open	0.40	1.73	mA	10
F2	A2			3					
F3	A3			2					
F4	A4			25					
F5	A5			26					
F6	A6			27					
F7	A7			7					
F8	A8			28					
F9	A9			9					
F10	A10			32					
F11	A11			13					
F12	A12			31					
F13	A13			34					
F14	A14			35					
F15	A15			17					
F16	A16			19					
F17	A17			39					
F18	A18	Input current, low level	Open	37	Open	0.40	1.73	mA	10

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TABLE IX. Input current (high-level) test.

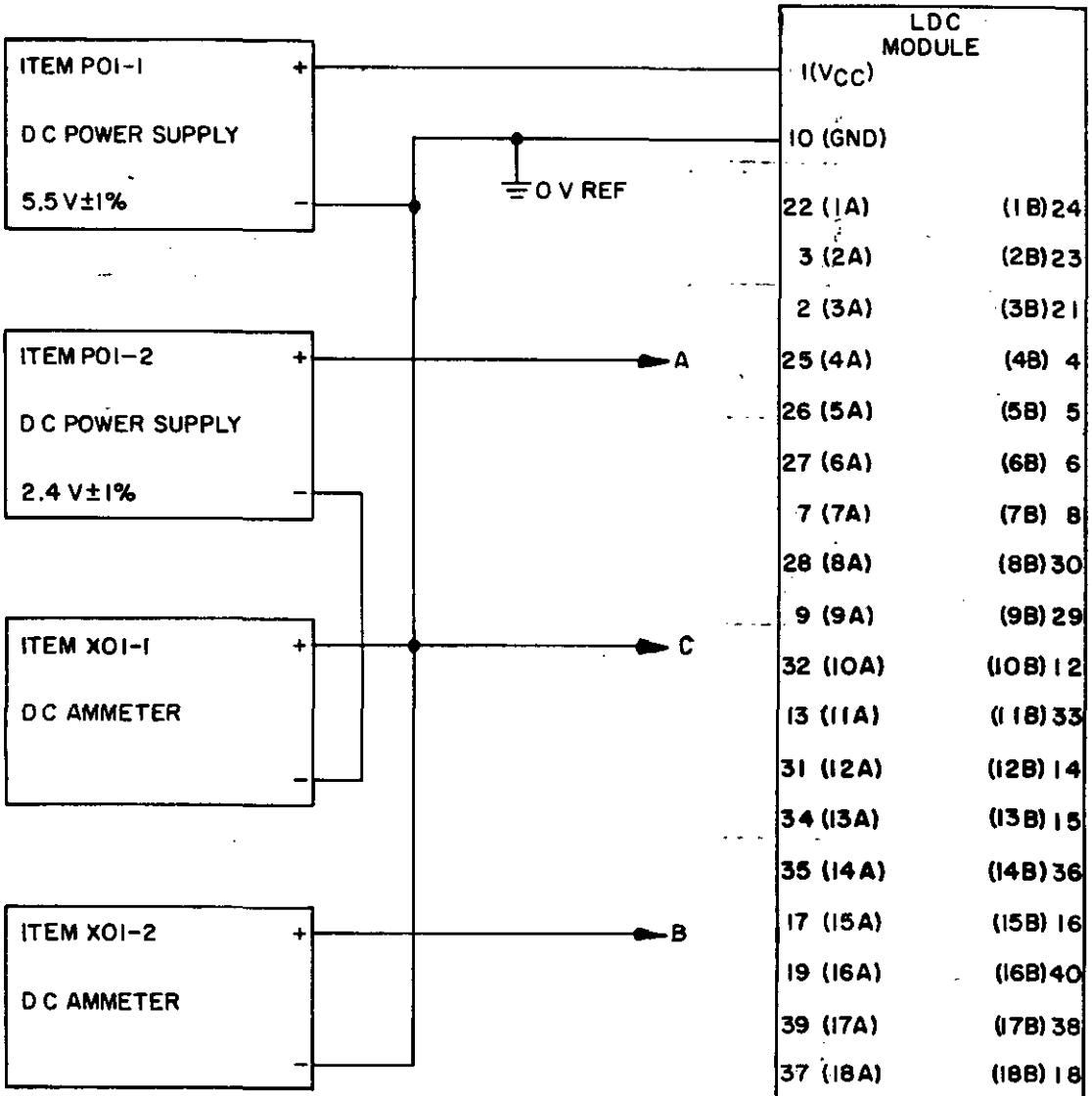
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	A1	Input current, high level	22	Open	Open	0	30	µA	10
G2	A2		3						
G3	A3		2						
G4	A4		25						
G5	A5		26						
G6	A6		27						
G7	A7		7						
G8	A8		28						
G9	A9		9						
G10	A10		32						
G11	A11		13						
G12	A12		31						
G13	A13		34						
G14	A14		35						
G15	A15		17						
G16	A16		19						
G17	A17		39						
G18	A18	Input current, high level	37	Open	Open	0	30	µA	10

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	B1 Output short circuit current	Open	24	22	20	55	mA	10
H2	B2		23	3				
H3	B3		21	2				
H4	B4		4	25				
H5	B5		5	26				
H6	B6		6	27				
H7	B7		8	7				
H8	B8		30	28				
H9	B9		29	9				
H10	B10		12	32				
H11	B11		33	13				
H12	B12		14	31				
H13	B13		15	34				
H14	B14		36	35				
H15	B15		16	17				
H16	B16		40	19				
H17	B17		38	39				
H18	B18 Output short circuit current	Open	18	37	20	55	mA	10

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-2 for tables VIII and X; use item X01-1 for table IX.
2. With point A disconnected, item X01-1 should indicate zero current on the 1 µA range.
3. When performing the output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

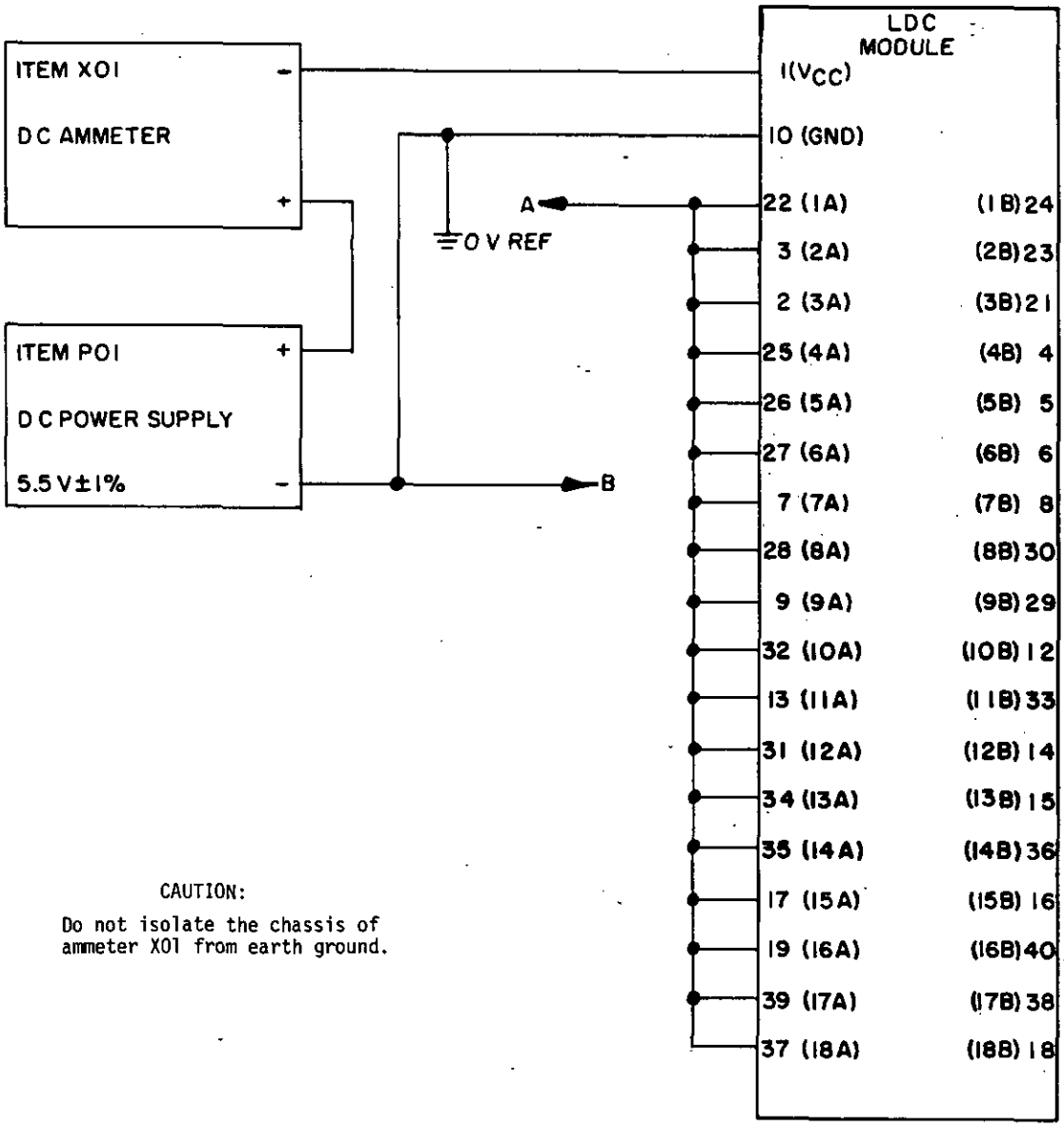
FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX, and X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Power supply current, low level	Open	Open	1	99	mA	11
J2	Power supply current, high level	B	A	1	36	mA	11

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 11. Power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH} and t_{THL}) test.

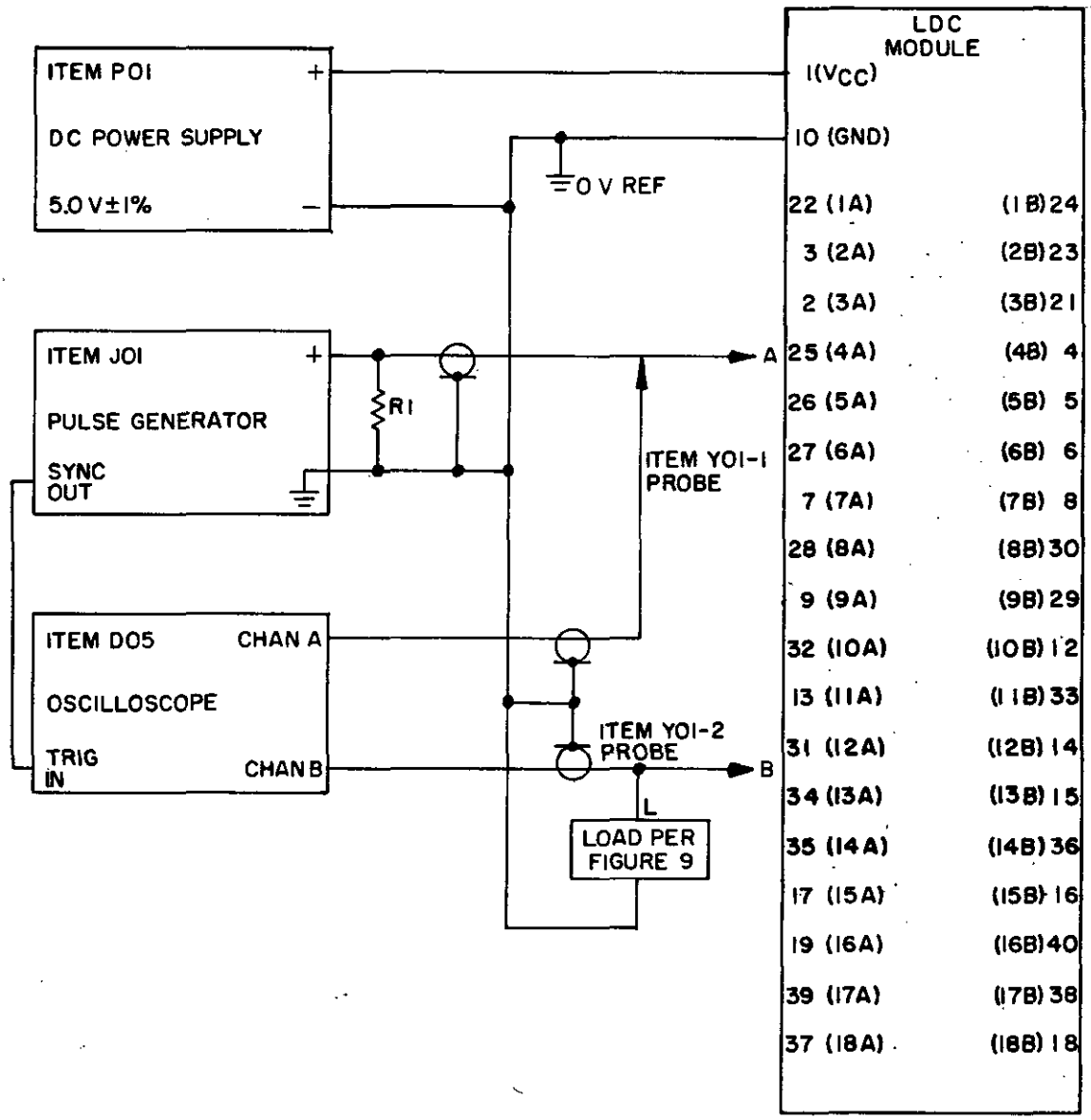
Test condition	Parameter		Connect		25° ±5°C ambient Initial limits				Unit	Test figure
			A to pin	B to pin	t_{TLH}		t_{THL}			
					Min	Max	Min	Max		
K1,L1	B1	Transition time	22	24	4	45	4	20	ns	12,13
K2,L2	B2		3	23						
K3,L3	B3		2	21						
K4,L4	B4		25	4						
K5,L5	B5		26	5						
K6,L6	B6		27	6						
K7,L7	B7		7	8						
K8,L8	B8		28	30						
K9,L9	B9		9	29						
K10,L10	B10		32	12						
K11,L11	B11		13	33						
K12,L12	B12		31	14						
K13,L13	B13		34	15						
K14,L14	B14		35	36						
K15,L15	B15		17	16						
K16,L16	B16		19	40						
K17,L17	B17		39	38						
K18,L18	B18	Transition time	37	18	4	45	4	20	ns	12,13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:29Z
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TABLE XIII. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter		Connect		25° ±5°C ambient Initial limits				Unit	Test figure
			A to pin	B to pin	t_{PHL}		t_{PLH}			
					Min	Max	Min	Max		
M1,N1	B1	Propagation delay time	22	24	2	19	4	28	ns	12,13
M2,N2	B2		3	23						
M3,N3	B3		2	21						
M4,N4	B4		25	4						
M5,N5	B5		26	5						
M6,N6	B6		27	6						
M7,N7	B7		7	8						
M8,N8	B8		28	30						
M9,N9	B9		9	29						
M10,N10	B10		32	12						
M11,N11	B11		13	33						
M12,N12	B12		31	14						
M13,N13	B13		34	15						
M14,N14	B14		35	36						
M15,N15	B15		17	16						
M16,N16	B16		19	40						
M17,N17	B17		39	38						
M18,N18	B18	Propagation delay time	37	18	2	19	4	28	ns	12,13

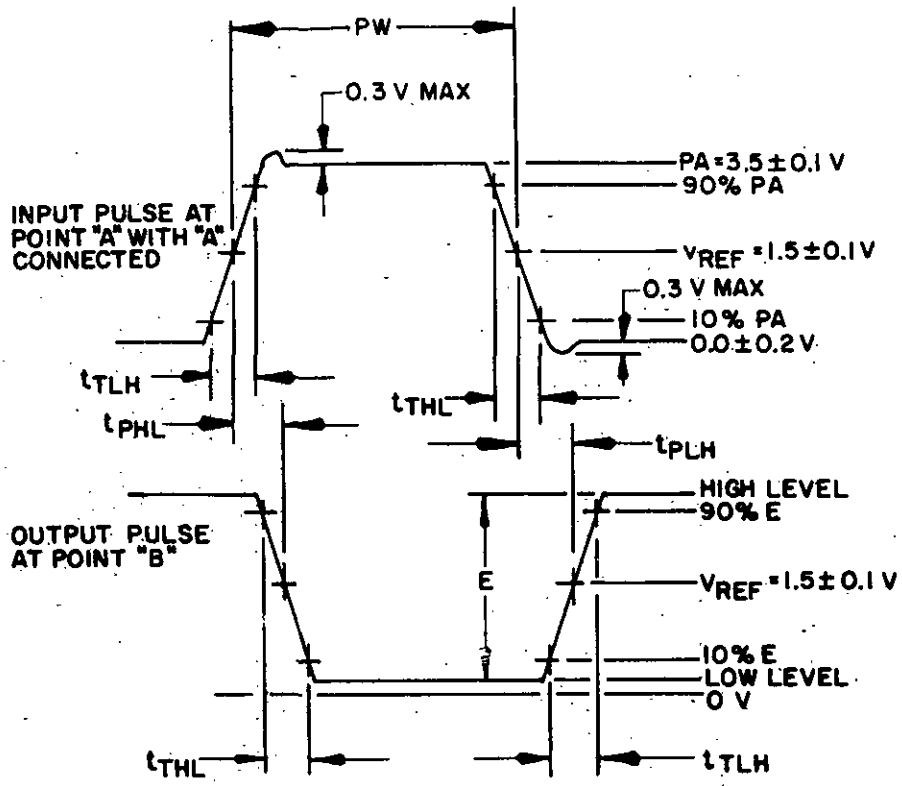
Source: https://assist.dla.mil -- Downloaded: 2026-04-20T15:29Z
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Before making measurements, the output levels shall be verified to be 0.4 V maximum for the low level and 2.4 V minimum for the high level.

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XII and XIII.

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NOTES:
1. Input pulse parameters:
   PW = 100 ns ±10%.
   PRR = 5 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltage levels are referenced to ground.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XII and XIII.

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TABLE XIV. Output crosstalk (low-level) test.

Test condition	Parameter		Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	B1	Output crosstalk, low level	Open	3,2,25,26,27,7,28,9,32,13,31, 34,35,17,19,39,37	22	24	0	970	mV	14,15
P2	B2			22,2,25,26,27,7,28,9,32,13, 31,34,35,17,19,39,37	3	23				
P3	B3			22,3,25,26,27,7,28,9,32,13, 31,34,35,17,19,39,37	2	21				
P4	B4			22,3,2,26,27,7,28,9,32,13,31, 34,35,17,19,39,37	25	4				
P5	B5			22,3,2,25,27,7,28,9,32,13,31, 34,35,17,19,39,37	26	5				
P6	B6			22,3,2,25,26,7,28,9,32,13,31, 34,35,17,19,39,37	27	6		970		
P7	B7			22,3,2,25,26,27,28,9,32,13, 31,34,35,17,19,39,37	7	8		870		
P8	B8			22,3,2,25,26,27,7,9,32,13,31, 34,35,17,19,39,37	28	30				
P9	B9			22,3,2,25,26,27,7,28,32,13, 31,34,35,17,19,39,37	9	29				
P10	B10			22,3,2,25,26,27,7,28,9,13,31, 34,35,17,19,39,37	32	12				
P11	B11			22,3,2,25,26,27,7,28,9,32,31, 34,35,17,19,39,37	13	33				
P12	B12	Output crosstalk, low level	Open	22,3,2,25,26,27,7,28,9,32,13, 34,35,17,19,39,37	31	14	0	870	mV	14,15

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TABLE XIV. Output crosstalk (low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P13	B13	Output crosstalk, low level	Open	22,3,2,25,26,27,7,28,9,32,13, 31,35,17,19,39,37	34	15	0	970	mV	14,15
P14	B14			22,3,2,25,26,27,7,28,9,32,13, 31,34,17,19,39,37	35	36				
P15	B15			22,3,2,25,26,27,7,28,9,32,13, 31,34,35,19,39,37	17	16				
P16	B16			22,3,2,25,26,27,7,28,9,32,13, 31,34,35,17,39,37	19	40				
P17	B17			22,3,2,25,26,27,7,28,9,32,13, 31,34,35,17,19,37	39	38				
P18	B18	Output crosstalk, low level	Open	22,3,2,25,26,27,7,28,9,32,13, 31,34,35,17,19,39	37	18	0	970	mV	14,15

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TABLE XIV. Output crosstalk (low-level) test - Continued.

Test condition		Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P19	B1	Output crosstalk, low level	Open	7,9,13,17,19,28,31, 32,34,35,37,39	2,3,22,25,26, 27	24	0	420	mV	14,15
P20	B2					23				
P21	B3					21				
P22	B4					4				
P23	B5					5				
P24	B6			7,9,13,17,19,28,31, 32,34,35,37,39	2,3,22,25,26, 27	6				
P25	B7			2,3,17,19,22,25,26, 27,34,35,37,39	7,9,13,28,31, 32	8				
P26	B8					30				
P27	B9					29				
P28	B10					12				
P29	B11					33				
P30	B12			2,3,17,19,22,25,26, 27,34,35,37,39	7,9,13,28,31, 32	14				
P31	B13	Output crosstalk, low level	Open	2,3,7,9,13,22,25,26, 27,28,31,32	17,19,34,35,37, 39	15	0	420	mV	14,15

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TABLE XIV. Output crosstalk (low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P32	B14	Output crosstalk, low level	Open	2,3,7,9,13,22,25,26, 27,28,31,32	17,19,34,35,37, 39	36	0	420	mV	14,15
P33	B15	↓	↓	↓	↓	16	↓	↓	↓	↓
P34	B16	↓	↓	↓	↓	40	↓	↓	↓	↓
P35	B17	↓	↓	↓	↓	38	↓	↓	↓	↓
P36	B18	Output crosstalk, low level	Open	2,3,7,9,13,22,25,26, 27,28,31,32	17,19,34,35,37, 39	18	0	420	mV	14,15

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TABLE XV. Output crosstalk (high-level) test.

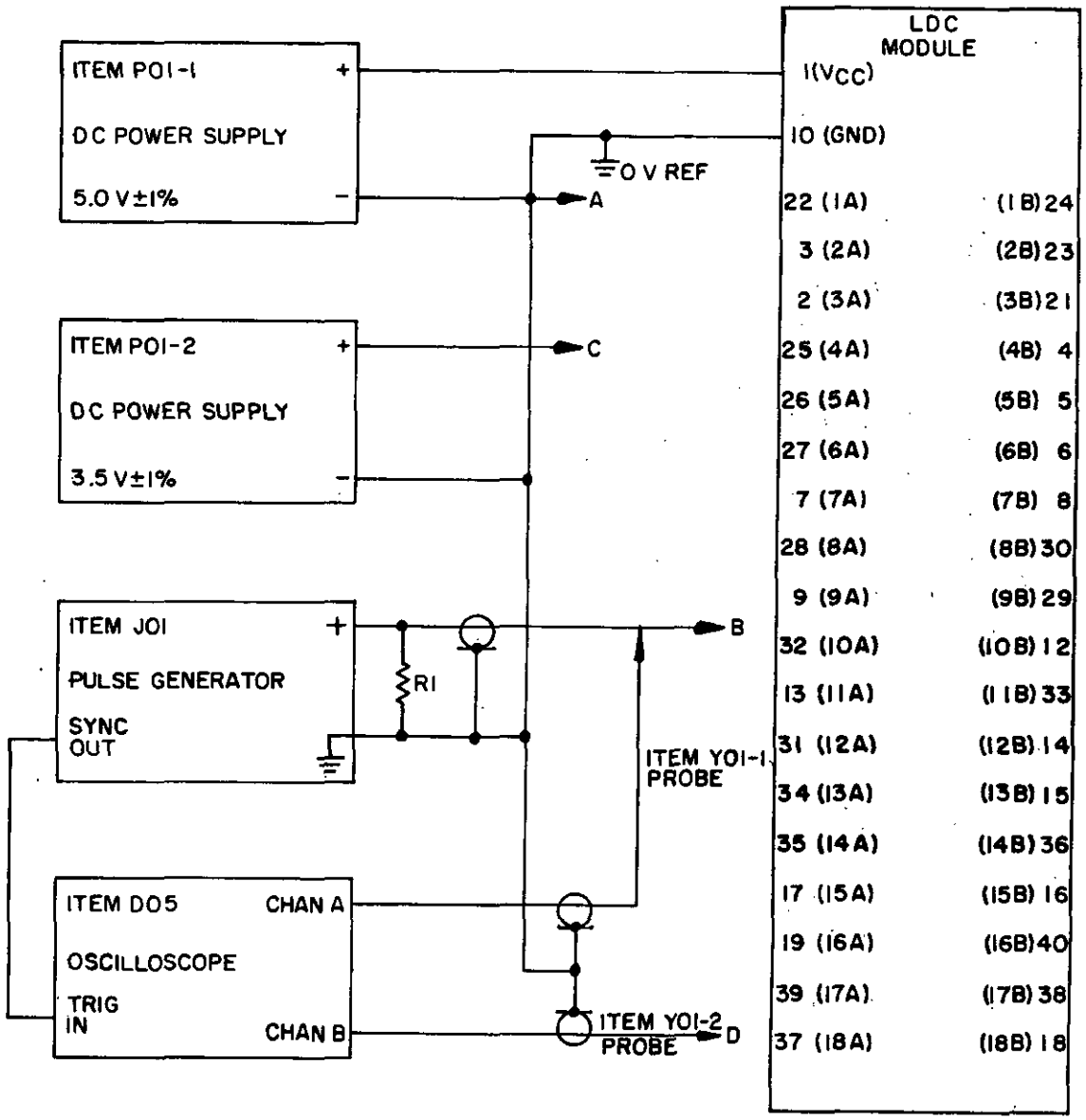
Test condition	Parameter		Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	B1	Output crosstalk, high level	22	3,2,25,26,27,7,28,9,32,13,31, 34,35,17,19,39,37	Open	24	0	400	mV	14,15
R2	B2		3	22,2,25,26,27,7,28,9,32,13,31, 34,35,17,19,39,37		23				
R3	B3		2	22,3,25,26,27,7,28,9,32,13,31, 34,35,17,19,39,37		21				
R4	B4		25	22,3,2,26,27,7,28,9,32,13,31, 34,35,17,19,39,37		4				
R5	B5		26	22,3,2,25,27,7,28,9,32,13,31, 34,35,17,19,39,37		5				
R6	B6		27	22,3,2,25,26,7,28,9,32,13,31, 34,35,17,19,39,37		6				
R7	B7		7	22,3,2,25,26,27,28,9,32,13,31, 34,35,17,19,39,37		8				
R8	B8		28	22,3,2,25,26,27,7,9,32,13,31, 34,35,17,19,39,37		30				
R9	B9		9	22,3,2,25,26,27,7,28,32,13,31, 34,35,17,19,39,37		29				
R10	B10		32	22,3,2,25,26,27,7,28,9,13,31, 34,35,17,19,39,37		12				
R11	B11		13	22,3,2,25,26,27,7,28,9,32,31, 34,35,17,19,39,37		33				
R12	B12	Output crosstalk, high level	31	22,3,2,25,26,27,7,28,9,32,13, 34,35,17,19,39,37	Open	14	0	400	mV	14,15

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TABLE XV. Output crosstalk (high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R13	B13	Output crosstalk, high level	34	22,3,2,25,26,27,7,28,9,32,13, 31,35,17,19,39,37	Open	15	0	400	mV	14,15
R14	B14	↓	35	22,3,2,25,26,27,7,28,9,32,13, 31,34,17,19,39,37	↓	36	↓	↓	↓	↓
R15	B15	↓	17	22,3,2,25,26,27,7,28,9,32,13, 31,34,35,19,39,37	↓	16	↓	↓	↓	↓
R16	B16	↓	19	22,3,2,25,26,27,7,28,9,32,13, 31,34,35,17,39,37	↓	40	↓	↓	↓	↓
R17	B17	↓	39	22,3,2,25,26,27,7,28,9,32,13, 31,34,35,17,19,37	↓	38	↓	↓	↓	↓
R18	B18	Output crosstalk, high level	37	22,3,2,25,26,27,7,28,9,32,13, 31,34,35,17,19,39	Open	18	0	400	mV	14,15

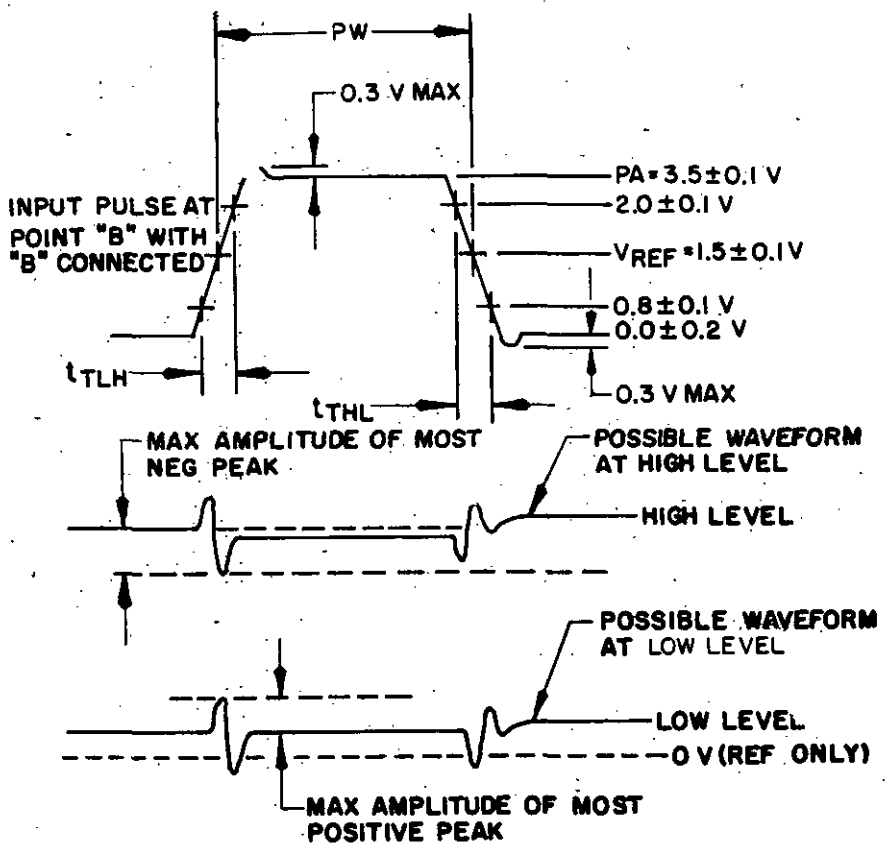
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load each output with the load specified on figure 9.

FIGURE 14. Output crosstalk (low-level and high-level) test circuit for tables XIV and XV.

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NOTES:
1. Input pulse parameters:
   PW = 200 ns ±10%
   PRR = 1 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
2. All voltage levels are referenced to ground.

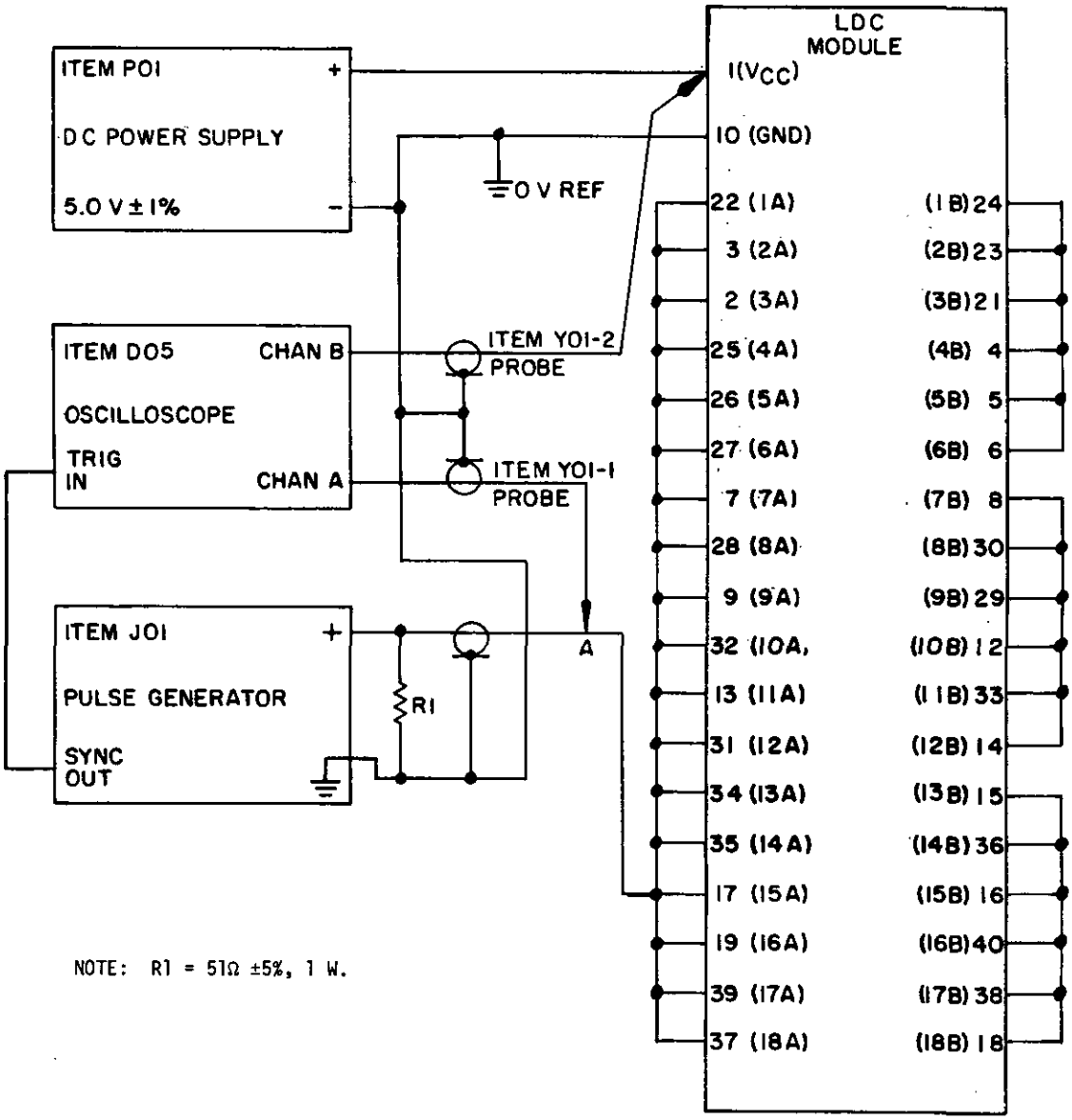
FIGURE 15. Output crosstalk (low-level and high-level) test circuit waveforms for tables XIV and XV.

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	600	mV	16, 17

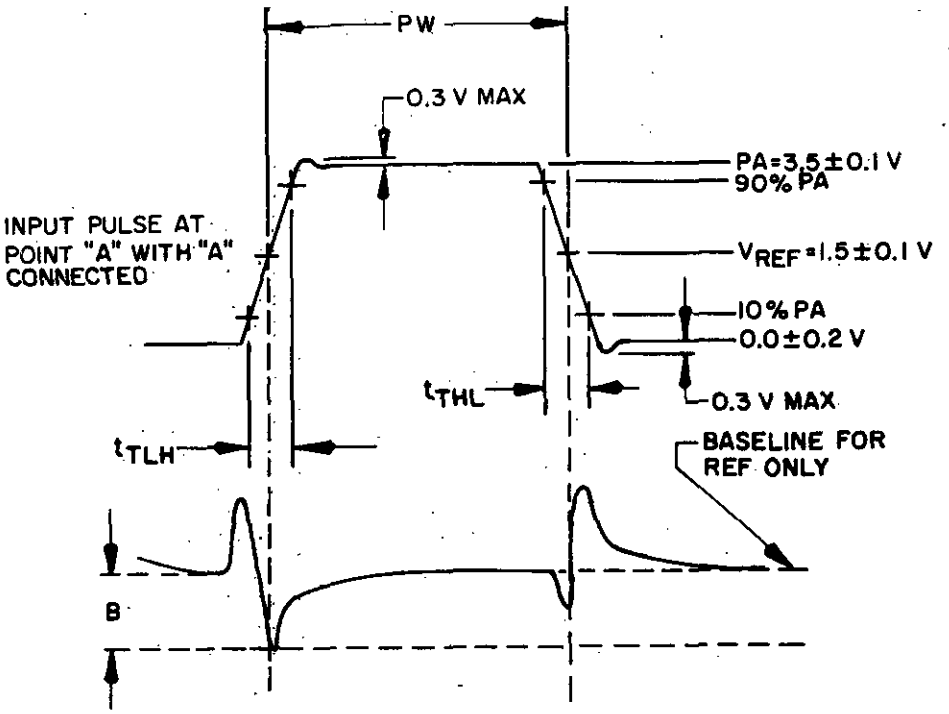
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 16. Power supply transient amplitude test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PW = 3 µs ±10%.
   PRR = 100 kHz ±10%.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns.
2. B = Measurement from the baseline to the most negative excursion of the waveform.
3. All voltage levels are referenced to ground.

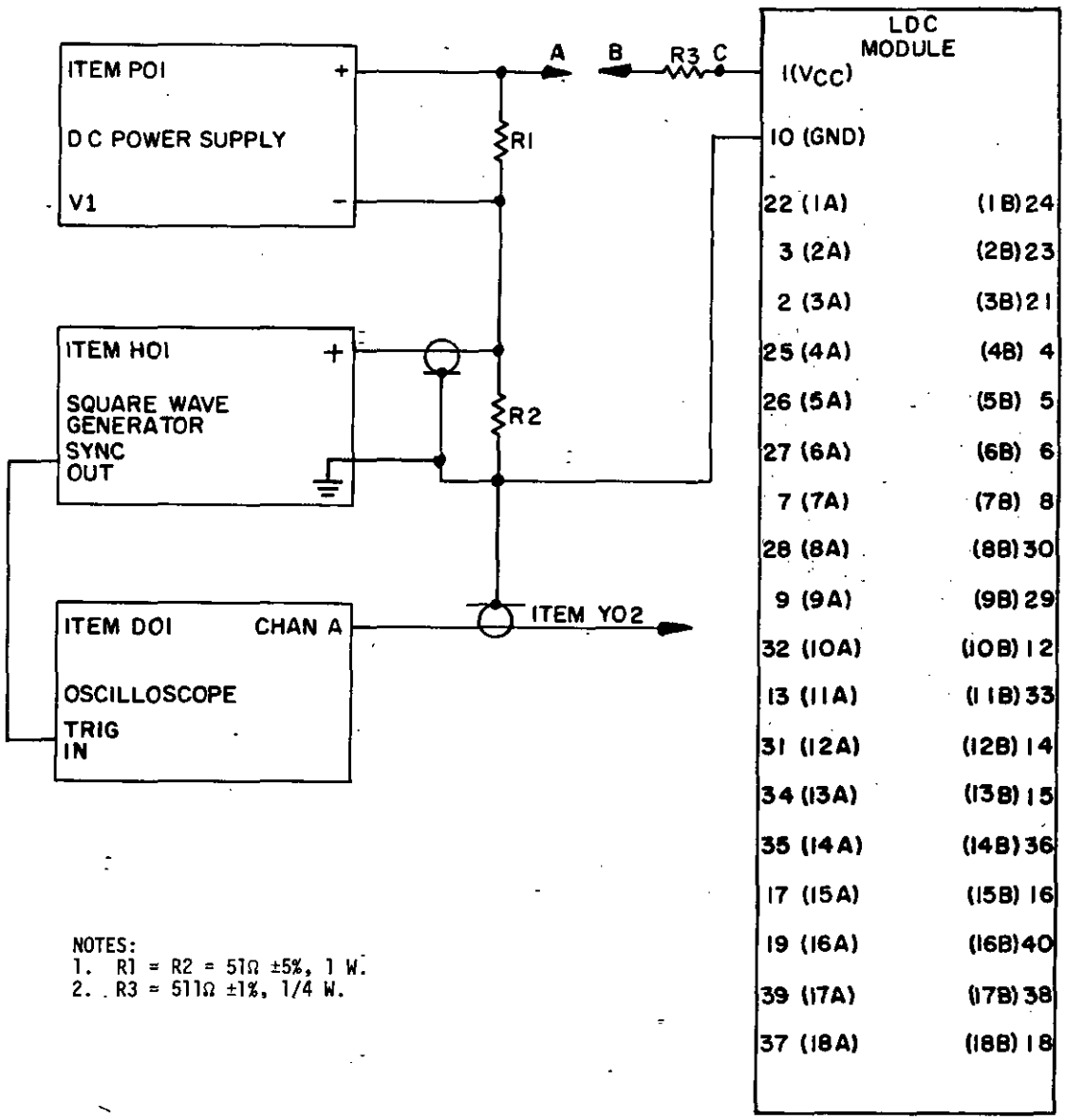
FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVI.

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TABLE XVII. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
T	Power supply transient rise time	1.0	2.0	ms	18, 19

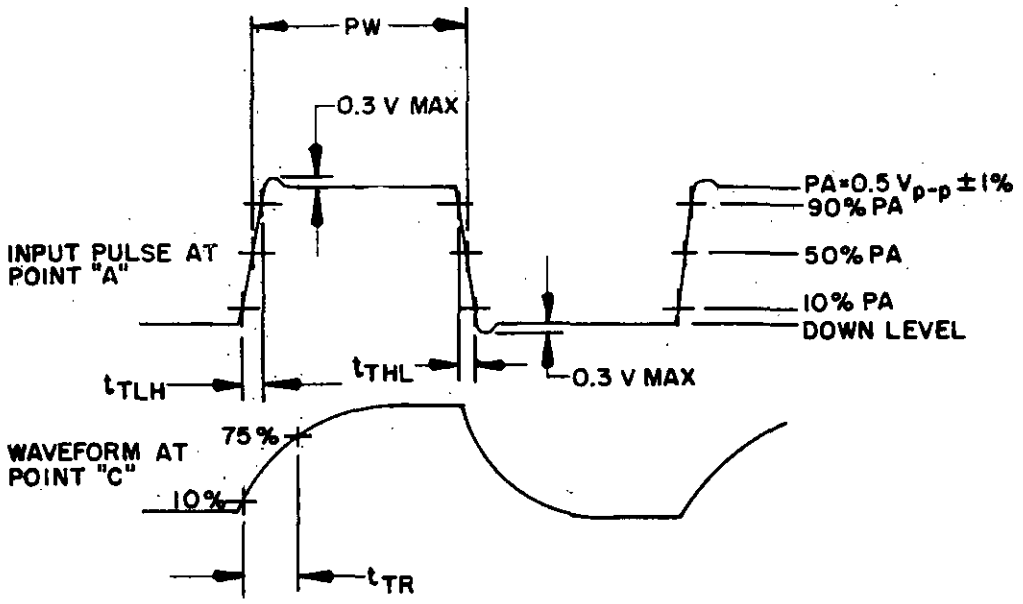
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 18. Power supply transient rise time test circuit waveforms for table XVII.

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NOTES:
1. Input pulse parameters:
   PW = 10 ms ±10%.
   PRR = 50 Hz (reference).
   $t_{TLH} = t_{THL} < 100$ ns (reference).
2. With point A open, adjust input shown.
3. With point A open, adjust V1 such that the down level of the input is at 0.00 ±0.01 V.
4. Connect A to B then recheck the down level and pulse parameters at point A.
5. Measure transient rise time (t_{TR}) at point C.

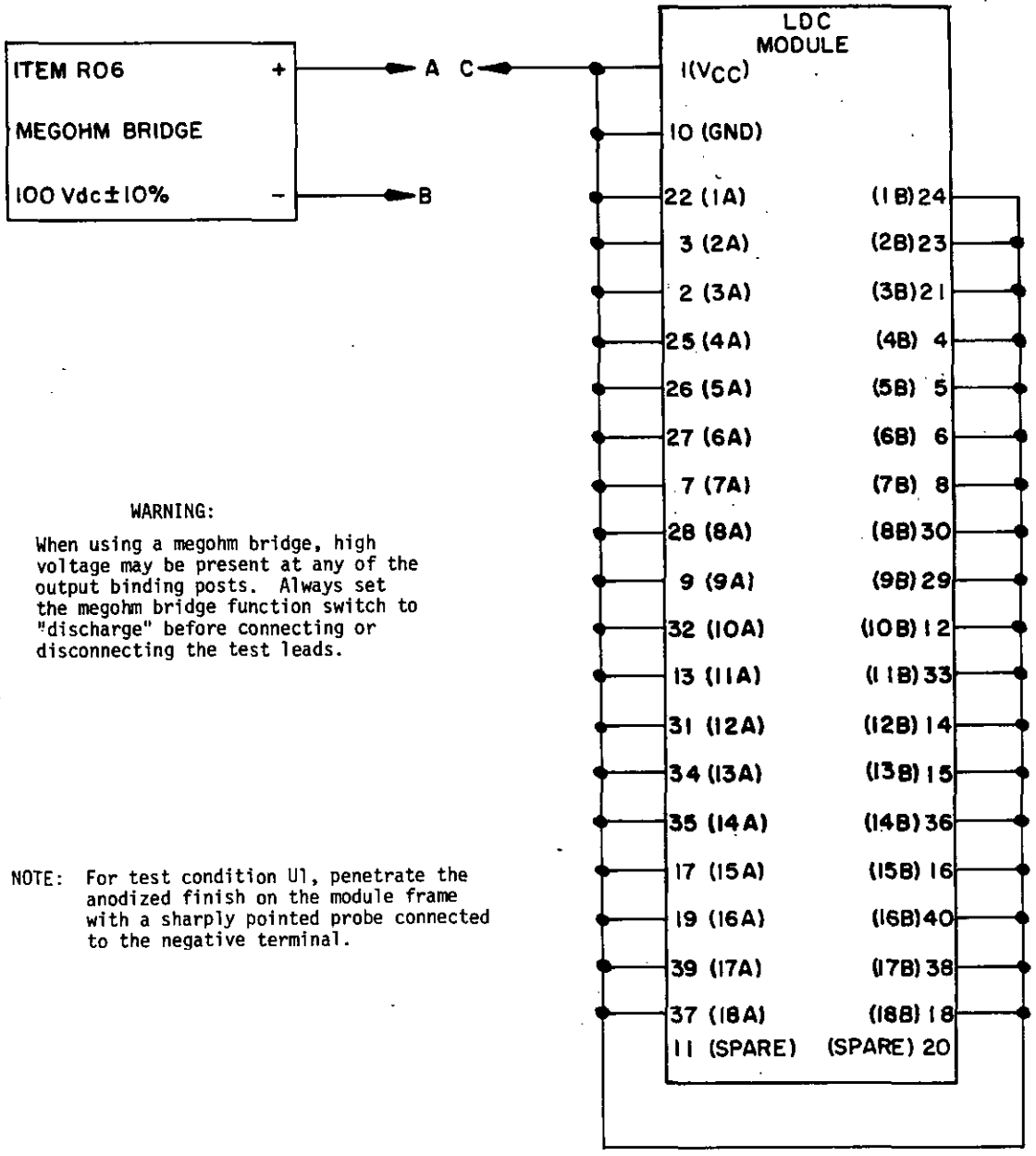
FIGURE 19. Power supply transient rise time test circuit waveforms for table XVII.

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TABLE XVIII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	Isolation	C,11,20	Frame	10	∞	MΩ	20
U2	Isolation, unused pins	C	11,20	10	∞	MΩ	20
U3	Isolation, unused pins	11	20	10	∞	MΩ	20

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition U1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

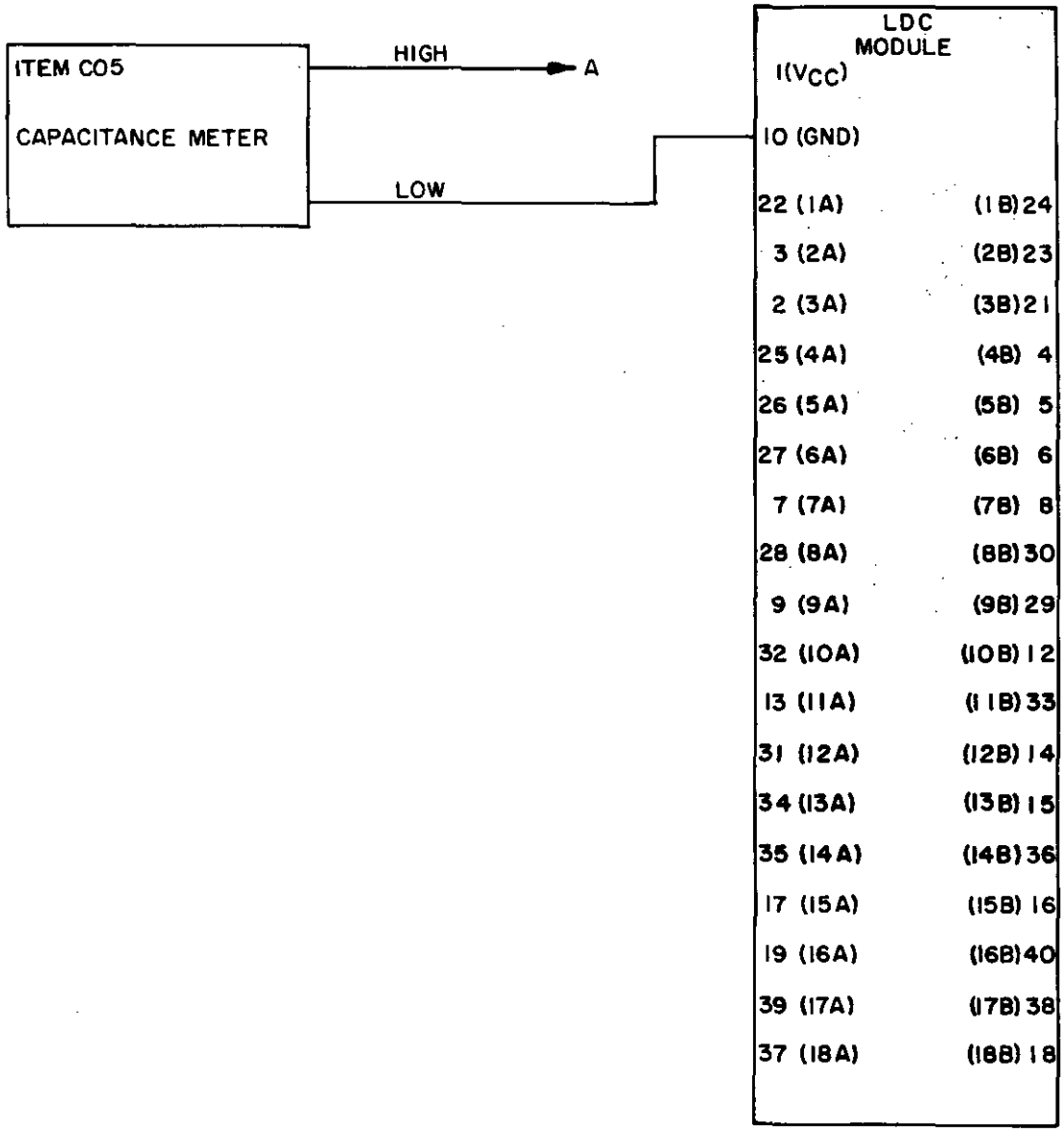
FIGURE 20. Isolation test circuit for table XVIII.

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TABLE XIX. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
V1	A1 Input capacitance	22	2	15	pF	21
V2	A2	3				
V3	A3	2				
V4	A4	25				
V5	A5	26				
V6	A6	27				
V7	A7	7				
V8	A8	28				
V9	A9	9				
V10	A10	32				
V11	A11	13				
V12	A12	31				
V13	A13	34				
V14	A14	35				
V15	A15	17				
V16	A16	19				
V17	A17	39				
V18	A18 Input capacitance	37	2	15	pF	21

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 21. Input capacitance test circuit for table XIX.

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TABLE XX. Output impedance, power off, check.

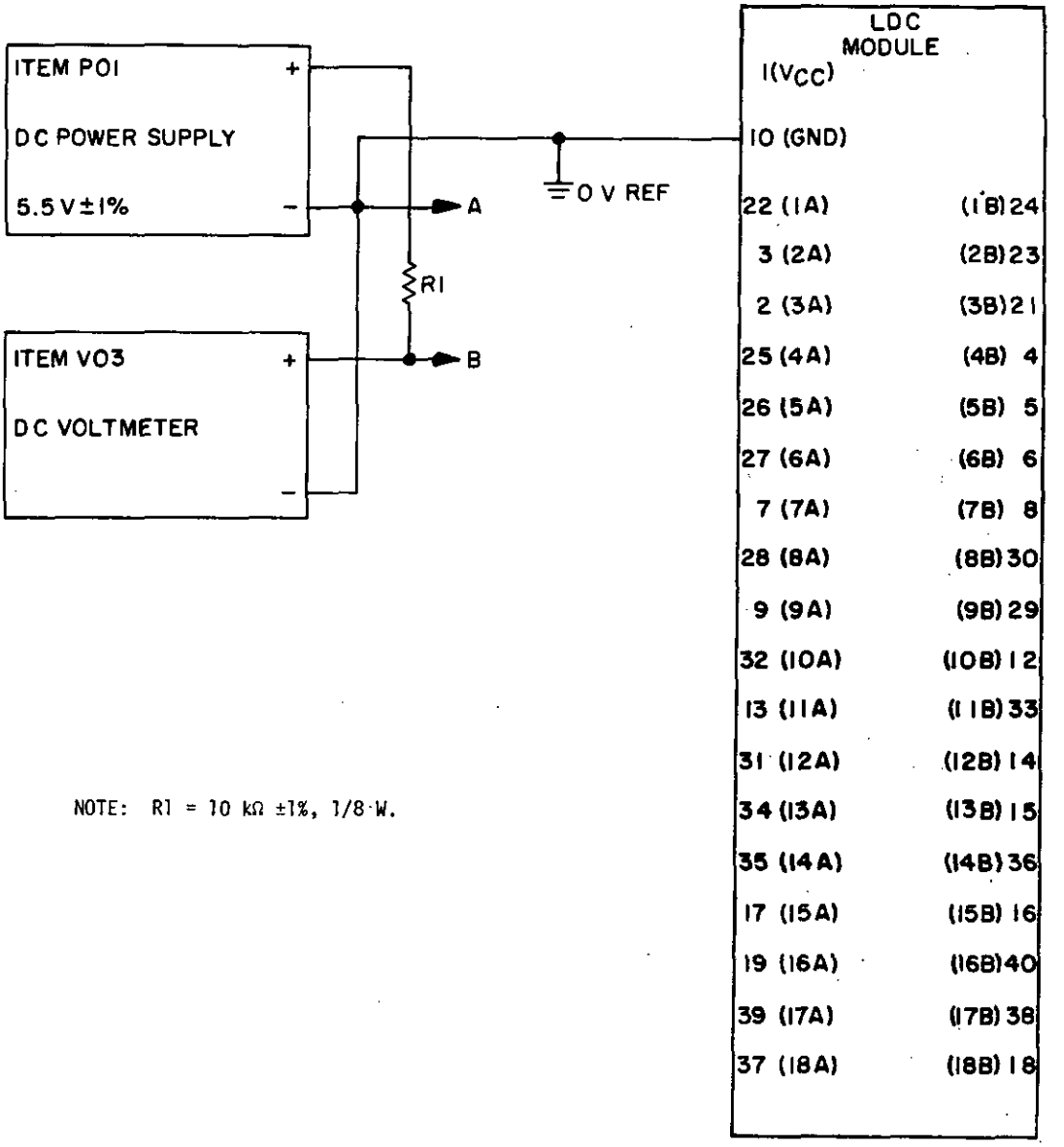
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W1	1B	Output impedance, power off, check	Open	24	3.5	5.55	V	22
W2	2B			23				
W3	3B			21				
W4	4B			4				
W5	5B			5				
W6	6B			6				
W7	7B			8				
W8	8B			30				
W9	9B			29				
W10	10B			12				
W11	11B			33				
W12	12B			14				
W13	13B			15				
W14	14B			36				
W15	15B			16				
W16	16B			40				
W17	17B			38				
W18	18B	Output impedance, power off, check	Open	18	3.5	5.55	V	22

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TABLE XX. Output impedance, power off, check - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W19	1B	Output impedance, power off, check	1	24	3.5	5.55	V	22
W20	2B			23				
W21	3B			21				
W22	4B			4				
W23	5B			5				
W24	6B			6				
W25	7B			8				
W26	8B			30				
W27	9B			29				
W28	10B			12				
W29	11B			33				
W30	12B			14				
W31	13B			15				
W32	14B			36				
W33	15B			16				
W34	16B			40				
W35	17B			38				
W36	18B	Output impedance, power off, check	1	18	3.5	5.55	V	22

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NOTE: R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 22. Output impedance, power off, check test circuit for table XX.

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